DI-OSVISE

Verification of Instruction Set Extensions Made Easy

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Motivation

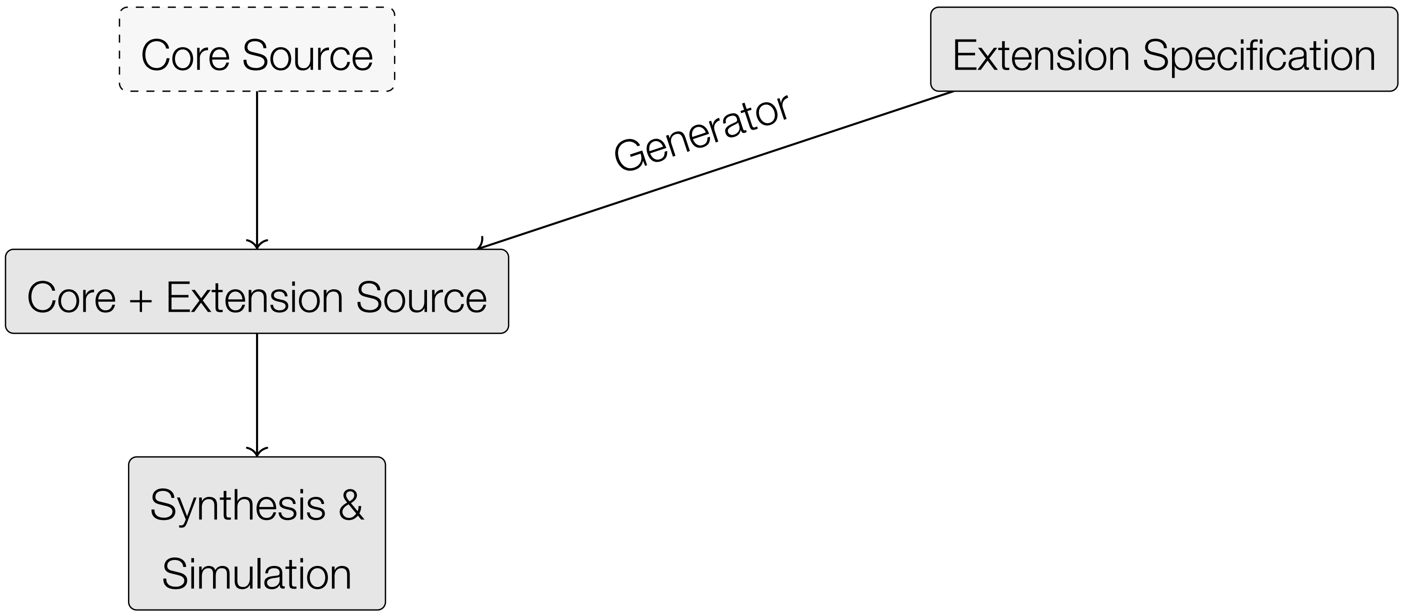
- RISC-V drives domain-specific instruction set extensions
- Instruction generators increase demand for verification

- Fast evaluation: generators, simulation
- Investigation of nonfunctional properties: performance, power
- Verification of new functionalities

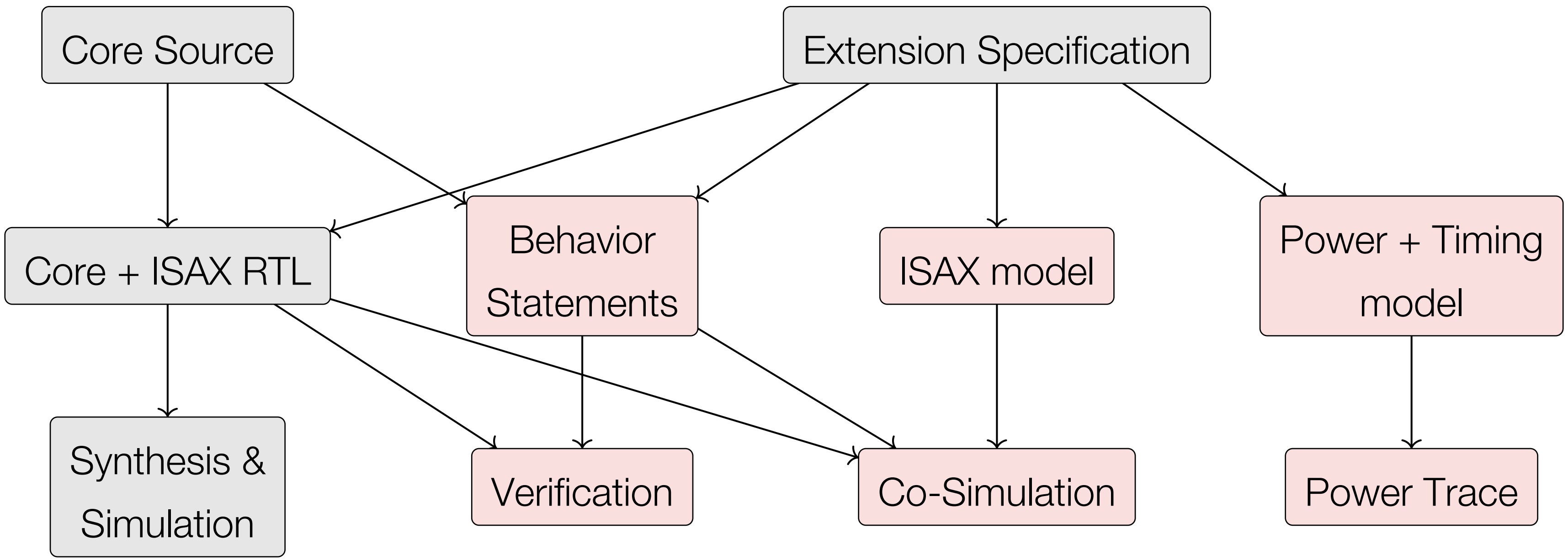
Open **S**ource **V**erification of **I**nstruction **S**et **E**xtensions

- New instruction set descriptions
- Extending verification capabilities
- Built on existing ecosystem
- Extend open-source tools

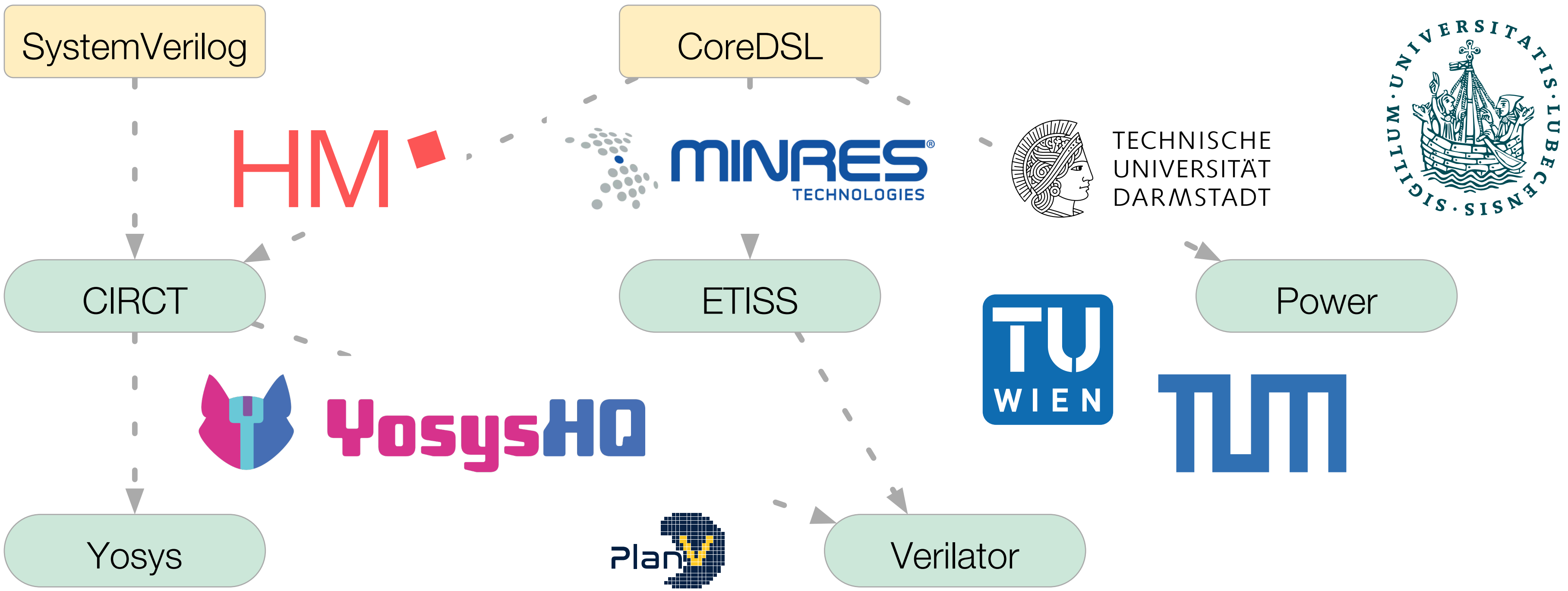
Improving Instruction Set Extension Flow



Improving Instruction Set Extension Flow

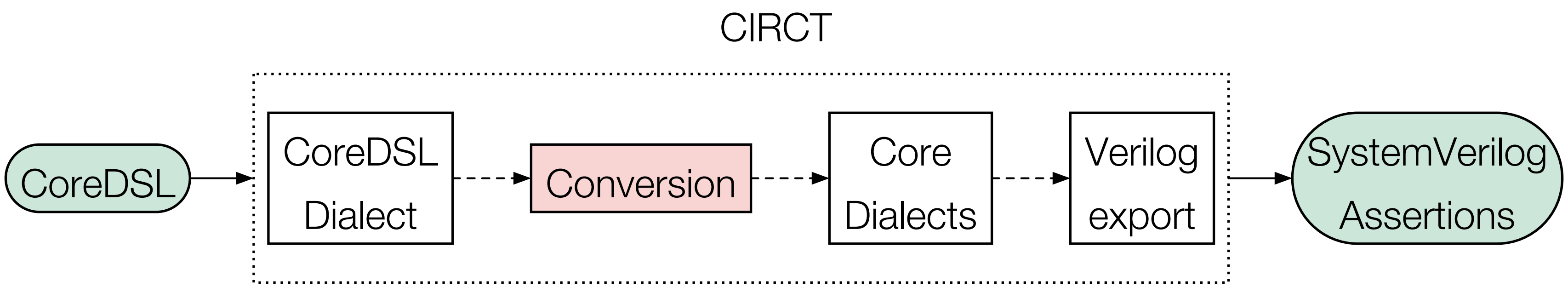


Tools and Project Partners



Verification Based on Open-Source Ecosystem

- **Objective**: Generate verification statements based on the instruction extension specification
- **Flow**: CoreDSL specification → SystemVerilog Assertions

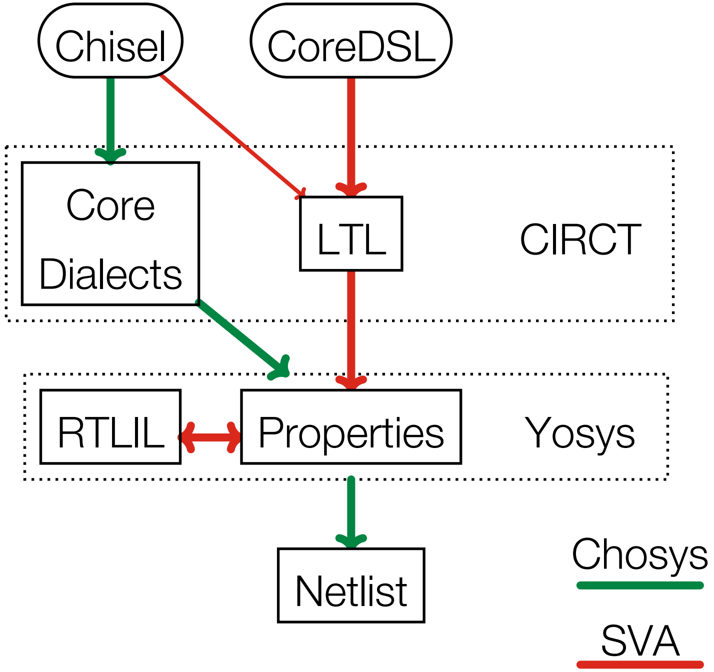


- **Reuse** of existing infrastructure: CoreDSL specification, CIRCT compiler
- Focus on new conversion pass to **extract** verification statements from behavior
- Open-source ecosystem facilitates complex flow and enables advanced automatic verification statements

Benefiting from Evolving Open-Source EDA

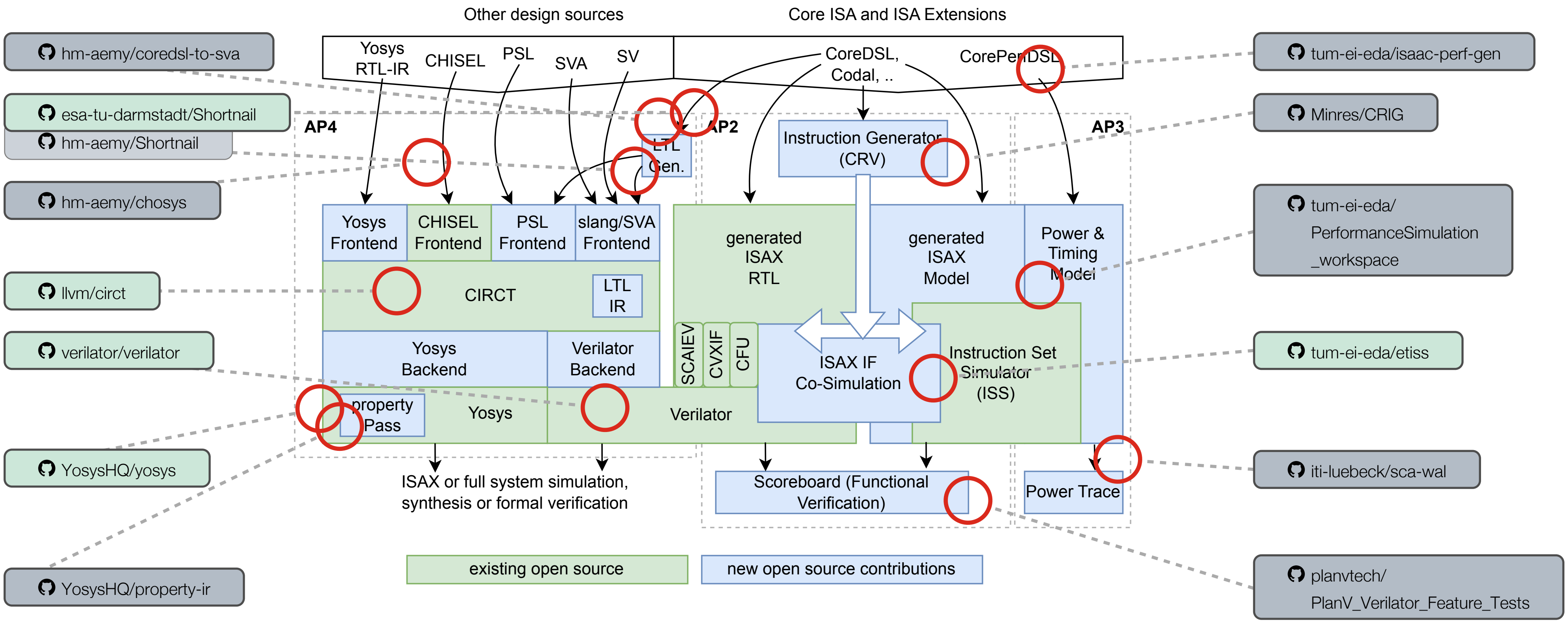
Leverage existing open-source ecosystem for complex and **novel** design flows.

- **Chosys**: Start with a Hardware Description Language (Chisel) and output a netlist.
  - No Verilog needed
  - Enables transportation of additional information
  - Focus on new dialects and passes
- Verification based on Linear Temporal Logic for extended SystemVerilog Assertions support



Extending existing ecosystem enables further advancements, allows for faster research and more capable tools.

Impact Open-Source Development



Impact Open-Source Development

- Development is done in the open
  - Instant feedback
  - Motivation for higher quality and reusability
- Enables outside collaboration
  - Higher impact by working with the community
- Continuous improvements and direct availability

Other design sources

Core ISA and ISA Extensions

hm-aemy/coredsl-to-sva

esa-tu-darmstadt/Shortnail

hm-aemy/Shortnail

hm-aemy/chosys

llvm/circt

verilator/verilator

YosysHQ/yosys

YosysHQ/property-ir

tum-ei-eda/isaac-perf-gen

Minres/CRIG

tum-ei-eda/PerformanceSimulation_workspace

tum-ei-eda/etiss

iti-luebeck/sca-wal

planvtech/PlanV_Verilator_Feature_Tests

synthesis or formal verification

Verification)

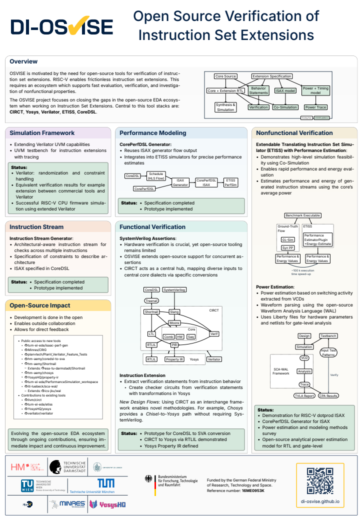
existing open source

new open source contributions

OSVISE Upcoming Conference Participation

Save the date for the leading open-source silicon conference

- **ORConf 2026**
  - September 11 to 13
  - Ghent, Belgium
- Meet the community
- Free to attend!
- Co-located with FPL



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Thank you!

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